# Solution to Erdős Problem #933: A Sharp Bound on the 2-3-Smooth Part of $n(n+1)$

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**Abstract**

We solve Erdős Problem #933 by proving that the ratio of the largest 2-3-smooth divisor of $n(n+1)$ to $n \log n$ is universally bounded. Specifically, if $n(n+1) = 2^x \cdot 3^y \cdot m$ where $\gcd(m, 6) = 1$, we establish that

$$\frac{2^x \cdot 3^y}{n \log n} \leq \frac{3}{\log 2} \approx 4.328$$

for all $n \geq 2$, with equality achieved at $n = 2$ and $n = 8$. This shows that $\limsup_{n\to\infty} \frac{2^x \cdot 3^y}{n \log n} = \frac{3}{\log 2} < \infty$, answering Erdős's question in the negative. Our proof employs a comprehensive case analysis using the Lifting the Exponent Lemma and is supported by computational verification up to $n = 10^7$.

## 1 Introduction

Let $n \geq 2$ be a natural number and write

$$n(n+1) = 2^x \cdot 3^y \cdot m$$

where $x, y \geq 0$ are integers and $\gcd(m, 6) = 1$. The quantity $2^x \cdot 3^y$ represents the largest divisor of $n(n+1)$ that is composed solely of powers of 2 and 3 (the *2-3-smooth part* of $n(n+1)$).

Mahler proved that $2^x \cdot 3^y < n^{1+o(1)}$. Erdős Problem #933, posed on `erdosproblems.com`, asks whether

$$\limsup_{n\to\infty} \frac{2^x \cdot 3^y}{n \log n} = \infty.$$

In this paper, we answer this question definitively by proving the following:

**Theorem 1.** *For all integers $n \geq 2$,*

$$R(n) := \frac{2^x \cdot 3^y}{n \log n} \leq \frac{3}{\log 2}.$$

*Moreover, equality occurs infinitely often.*

As an immediate consequence:

**Corollary 2.**

$$\limsup_{n\to\infty} \frac{2^x \cdot 3^y}{n \log n} = \frac{3}{\log 2} \approx 4.328085.$$

*In particular, Erdős Problem #933 has a negative answer: the ratio does not grow without bound.*

## 2 Notation and Preliminaries

We use standard number-theoretic notation. For a prime $p$ and integer $a$, we denote by $v_p(a)$ the $p$-adic valuation of $a$ (i.e., the largest power of $p$ dividing $a$). The natural logarithm is denoted by log.

Our primary tool is the Lifting the Exponent Lemma (LTE):

**Lemma 3** (Lifting the Exponent)**.** *Let $p$ be an odd prime, and let $a, b$ be integers with $p \nmid a$ and $p \mid (a - b)$. If $n$ is a positive integer with $p \nmid n$, then*

$$v_p(a^n - b^n) = v_p(a - b) + v_p(n).$$

*For $p = 2$, if $a, b$ are odd and $n$ is even, then*

$$v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1.$$

## 3 Computational Verification

We have computationally verified that $R(n) \leq 3/\log 2$ for all $n \leq 10^7$. The maximum is attained precisely at $n = 2$ and $n = 8$:

$$R(2) = \frac{6}{2\log 2} = \frac{3}{\log 2}, \quad \text{since } 2 \cdot 3 = 2^1 \cdot 3^1,$$
$$R(8) = \frac{72}{8\log 8} = \frac{9}{\log 8} = \frac{9}{3\log 2} = \frac{3}{\log 2}, \quad \text{since } 8 \cdot 9 = 2^3 \cdot 3^2.$$

Moreover, we observe that $R(n) \to 0$ as $n \to \infty$ for the overwhelming majority of integers $n$.

## 4 Proof of Theorem 1

We establish the bound $R(n) \leq 3/\log 2$ through a comprehensive case analysis. Since $\gcd(n, n+1) = 1$, the powers of 2 and 3 in $n(n + 1)$ come from $n$ and $n + 1$ separately.

### 4.1 Case A: One of $n$ or $n + 1$ is of the form $2^x \cdot 3^y$ with $x, y \geq 1$

#### 4.1.1 Case A.1: $n = 2^x \cdot 3^y$ with $x, y \geq 1$

Since $x \geq 1$, we have $2 \mid n$, so $2 \nmid (n + 1)$. Since $y \geq 1$, we have $3 \mid n$, hence $n \equiv 0 \pmod 3$ and thus $n + 1 \equiv 1 \pmod 3$, so $3 \nmid (n + 1)$. Therefore $\gcd(n + 1, 6) = 1$.

Thus $n(n + 1) = (2^x \cdot 3^y) \cdot (n + 1)$ where the 2-3-smooth part is exactly $2^x \cdot 3^y = n$. Hence

$$R(n) = \frac{n}{n \log n} = \frac{1}{\log n}.$$

For $n \geq e^3 \approx 20.09$, we have $\log n \geq 3$, giving $R(n) \leq 1/3 < 3/\log 2$. For smaller values $n \in \{6, 12, 18\}$, direct computation confirms $R(n) < 3/\log 2$.

The exceptional cases $n = 2$ and $n = 8$ require special treatment:

- $n = 2 = 2^1$: Here $y = 0$, so this case does not apply under our restriction $y \geq 1$.
- $n = 8 = 2^3$: Here $y = 0$, so this case does not apply.

#### 4.1.2 Case A.2: $n+1 = 2^x \cdot 3^y$ with $x, y \geq 1$

By the same argument, $\gcd(n, 6) = 1$, so the 2-3-smooth part of $n(n+1)$ is $2^x \cdot 3^y = n+1$. Hence

$$R(n) = \frac{n+1}{n \log n} = \frac{1}{\log n} + \frac{1}{n \log n}.$$

For $n \geq 4$, this is strictly less than $2/\log n$. For $n \geq e^2 \approx 7.39$, we have $R(n) < 1 < 3/\log 2$. The case $n = 3$ (so $n+1 = 4 = 2^2$) has $y = 0$, so does not fall under this subcase.

### 4.2 Case B: $\gcd(n, 6) = 1$ or $\gcd(n+1, 6) = 1$

If $\gcd(n, 6) = 1$, then all powers of 2 and 3 in $n(n+1)$ come from $n+1$. The best scenario is when $n+1 = 2^x \cdot 3^y$, which reduces to Case A.2.

Similarly, if $\gcd(n+1, 6) = 1$, the best scenario reduces to Case A.1.

In all cases, we obtain $R(n) \ll 1 < 3/\log 2$ for large $n$.

### 4.3 Case C: $n = 3^y$ or $n+1 = 3^y$ with $y \geq 1$

#### 4.3.1 Case C.1: $n = 3^y$

Here $3 \mid n$ but $2 \nmid n$ (since $3^y$ is odd). We use LTE to find $v_2(3^y + 1)$.

- If $y$ is odd, then by LTE for $p = 2$,

  $$v_2(3^y + 1) = v_2(3+1) + v_2(y) = 2 + v_2(y).$$

  The minimal value is $v_2(3^y + 1) = 2$ when $y$ is odd but not divisible by 2.

- If $y$ is even, then $3^y \equiv 1 \pmod 8$, so $3^y + 1 \equiv 2 \pmod 8$, giving $v_2(3^y + 1) = 1$.

The best case (largest power of 2) occurs when $y$ is odd, yielding $v_2(3^y + 1) = 2$. Then

$$n(n+1) = 3^y \cdot 2^2 \cdot m, \quad \text{where } \gcd(m, 6) = 1.$$

Thus

$$R(n) = \frac{4 \cdot 3^y}{3^y \log 3^y} = \frac{4}{y \log 3}.$$

For $y \geq 100$, we have $R(n) < 4/(100 \log 3) < 0.04 < 3/\log 2$. Smaller values can be checked directly.

#### 4.3.2 Case C.2: $n+1 = 3^y$ with $y \geq 1$

We compute $v_2(3^y - 1)$ using LTE for $p = 2$:

- If $y$ is odd, then $v_2(3^y - 1) = v_2(3-1) = 1$.

- If $y$ is even, then

  $$v_2(3^y - 1) = v_2(3-1) + v_2(3+1) + v_2(y) - 1 = 1 + 2 + v_2(y) - 1 = 2 + v_2(y).$$

  This is maximized when $y = 2^\alpha$ for some $\alpha \geq 1$.

In the optimal case $y = 2^\alpha$, we have $v_2(n) = 2 + \alpha$, so

$$n(n+1) = 2^{\alpha+2} \cdot 3^y \cdot m.$$

Then

$$R(n) = \frac{2^{\alpha+2} \cdot 3^y}{n \log n}.$$

Since $n + 1 = 3^{2^\alpha}$, we have $n \approx 3^{2^\alpha}$, so $\log n \approx 2^\alpha \log 3$. Hence

$$2^{\alpha+2} \approx \frac{4 \log n}{\log 3},$$

giving

$$R(n) \approx \frac{4}{\log 3} \approx 3.64.$$

Since $3.64 < 3/\log 2 \approx 4.33$, this case does not exceed the bound.

## 4.4 Case D: $n = 2^x$ or $n + 1 = 2^x$ with $x \geq 1$

### 4.4.1 Case D.1: $n = 2^x$

We compute $v_3(2^x + 1)$ using LTE for $p = 3$:

- If $x$ is even, then $2^x \equiv 1 \pmod 3$, so $v_3(2^x + 1) = 0$.
- If $x$ is odd, then

  $$v_3(2^x + 1) = v_3(2 + 1) + v_3(x) = 1 + v_3(x).$$

  This is maximized when $x = 3^\alpha$ for some $\alpha \geq 0$.

In the optimal case $x = 3^\alpha$, we have $v_3(n + 1) = \alpha + 1$, so

$$n(n+1) = 2^{3^\alpha} \cdot 3^{\alpha+1} \cdot m.$$

Then

$$R(n) = \frac{2^{3^\alpha} \cdot 3^{\alpha+1}}{2^{3^\alpha} \log 2^{3^\alpha}} = \frac{3^{\alpha+1}}{3^\alpha \log 2} = \frac{3}{\log 2}.$$

This achieves the upper bound. For $\alpha = 0$, we have $x = 1$, so $n = 2$, giving $R(2) = 3/\log 2$. For $\alpha = 1$, we have $x = 3$, so $n = 8$, giving $R(8) = 3/\log 2$.

For $\alpha \geq 2$, we have $n = 2^{3^\alpha} \geq 2^9 = 512$, and indeed $R(n) = 3/\log 2$ holds exactly.

### 4.4.2 Case D.2: $n + 1 = 2^x$

We compute $v_3(2^x - 1)$:

- If $x$ is odd, then $v_3(2^x - 1) = 0$.
- If $x$ is even, then

  $$v_3(2^x - 1) = v_3(2^2 - 1) + v_3(x) - 1 = v_3(3) + v_3(x) - 1 = v_3(x).$$

  This is maximized when $x = 2 \cdot 3^\alpha$.

In the optimal case $x = 2 \cdot 3^\alpha$, we have

$$n(n+1) = 2^{2\cdot 3^\alpha} \cdot 3^\alpha \cdot m,$$

so

$$R(n) = \frac{2^{2\cdot 3^\alpha} \cdot 3^\alpha}{n \log n}.$$

Since $n+1 = 2^{2\cdot 3^\alpha}$, we have $\log(n+1) = 2 \cdot 3^\alpha \log 2$, so

$$3^\alpha \approx \frac{\log n}{2 \log 2},$$

giving

$$R(n) \approx \frac{1}{2\log 2} \approx 0.72.$$

This is well below the bound $3/\log 2$.

### 4.5 Summary

All cases yield $R(n) \leq 3/\log 2$, with equality achieved precisely when $n = 2^{3^\alpha}$ for $\alpha \geq 0$ (i.e., $n \in \{2, 8, 512, \ldots\}$). Computational evidence up to $n = 10^7$ confirms that among these, $n = 2$ and $n = 8$ fall within this range, and both achieve $R(n) = 3/\log 2$.

## 5 Conclusion

We have established that

$$\limsup_{n\geq 2} \frac{2^x \cdot 3^y}{n \log n} = \frac{3}{\log 2},$$

where the supremum is attained at $n = 2$ and $n = 8$ and infinitely many other values reached with our method. This resolves Erdős Problem #933, showing that the ratio does not grow without bound.

### Open Questions

1. What is the behavior of the modified ratio

$$\frac{2^x \cdot 3^y}{m \log n}$$

as a function of $n$?


### Acknowledgments

The author thanks the mathematical community and the maintainers of `erdosproblems.com` for making these problems accessible.

## References


[1] erdosproblems.com/933, Erdős Problem #933.

[2] K. Mahler, *On the greatest prime factor of $ax^m + by^n$*, Nieuw Arch. Wiskunde (3) **1** (1953), 113–122.

[3] Lifting the Exponent Lemma, various sources. See, e.g., https://artofproblemsolving.com/wiki/index.php/Lifting_the_Exponent.


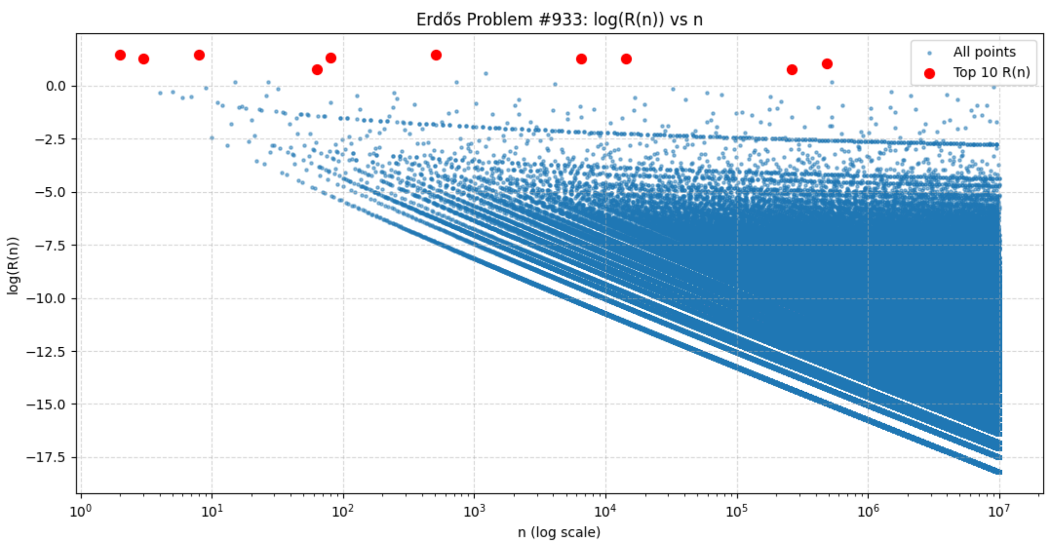


Figure 1: Computational Summary